Team Name: Antriksha

Team Members:

1. Aditya Krishna Jaiswal
2. Pratham Agarwal
3. Krish Kahnani
4. Keshav Jindal

Detailed solution and Approach

The proposed solution enhances satellite image analysis by integrating an Object-aware Refinement method into a multi-task deep learning architecture. This architecture leverages four key components:

1. **Mobile Net**: Serves as the shared feature extractor, providing rich, multi-class semantic segmentation for all subsequent tasks.

Why use it? Computationally faster and less intensive with comparable results with heavier CNN models like Resnet50 and Mask RCNN.

2. **YOLO Head:** performs object detection, outputting bounding boxes, object classes, and confidence scores.

Why use it?

State of the art open source algorithm used industry wide for object detection in robotics and autonomous systems.

Proposed architecture/user diagram.

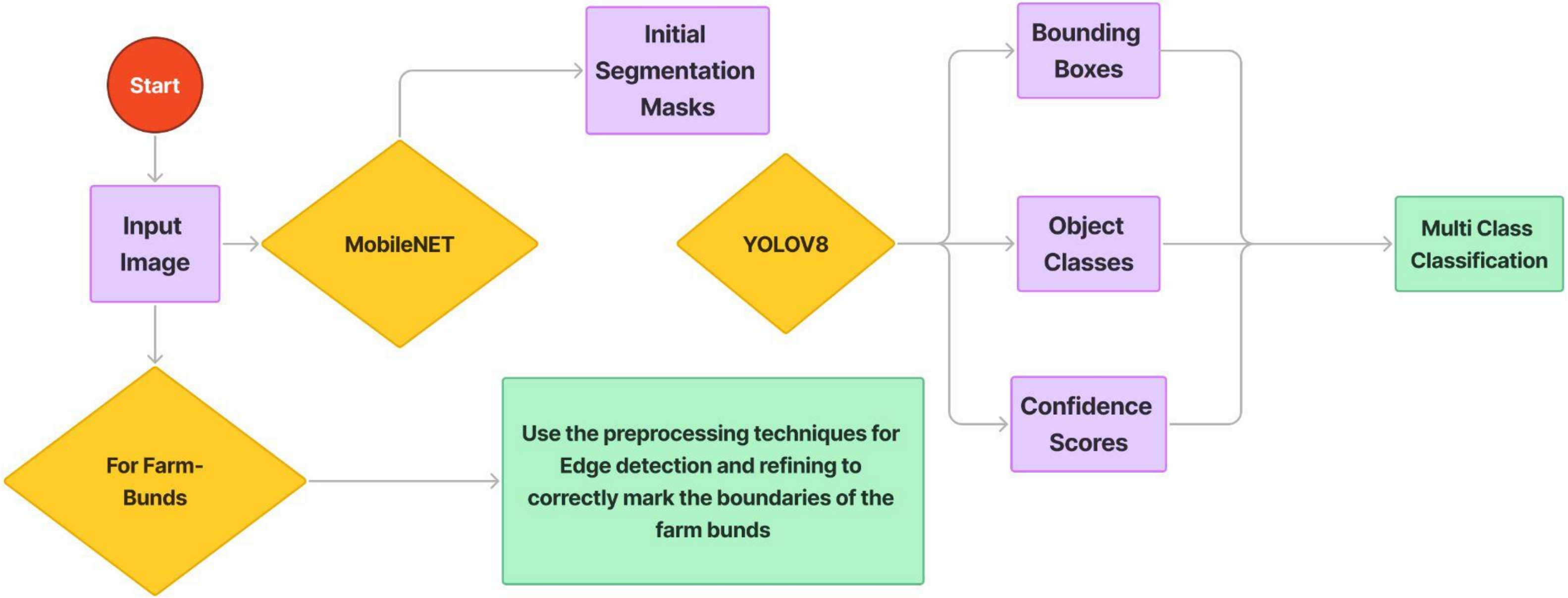


Figure: Proposed Solution Flow Diagram

How will it be able to solve the problem?

Our proposed solution addresses the problem by:

- **Multi-model integration:** Combining powerful feature extraction , and YOLO's robust object detection, the solution captures a more comprehensive understanding of the satellite imagery.

- **Class-specific strategies**: By applying tailored refinement rules for different object types (e.g. farm bunds etc.) the solution better handles the diverse elements in satellite imagery.

Tools and Technology Used:

For Preprocessing:

- **QGIS (v3.28)** - Used for labeling satellite images with image-level tags, bounding boxes, and masks.
- **GDAL (v3.5.0)** - Library for handling and processing geospatial data formats, essential for satellite imagery preprocessing.
- **OpenCV:-** For preparing training data (applying masks,segmentation etc.)

For model development and training :

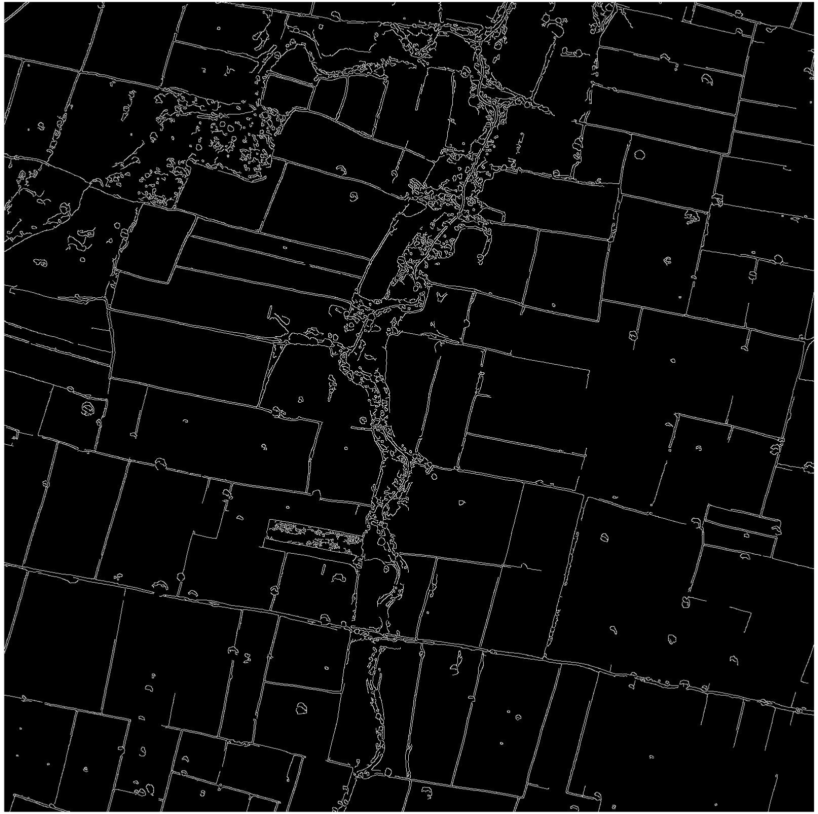
- **Python libraries(v3.11)** - Core programming language for the entire project, including data processing, model development, deployment and post processing.
- **PyTorch (v2.40)** - Deep learning frameworks for developing models for multi-label classification, localization, and segmentation.

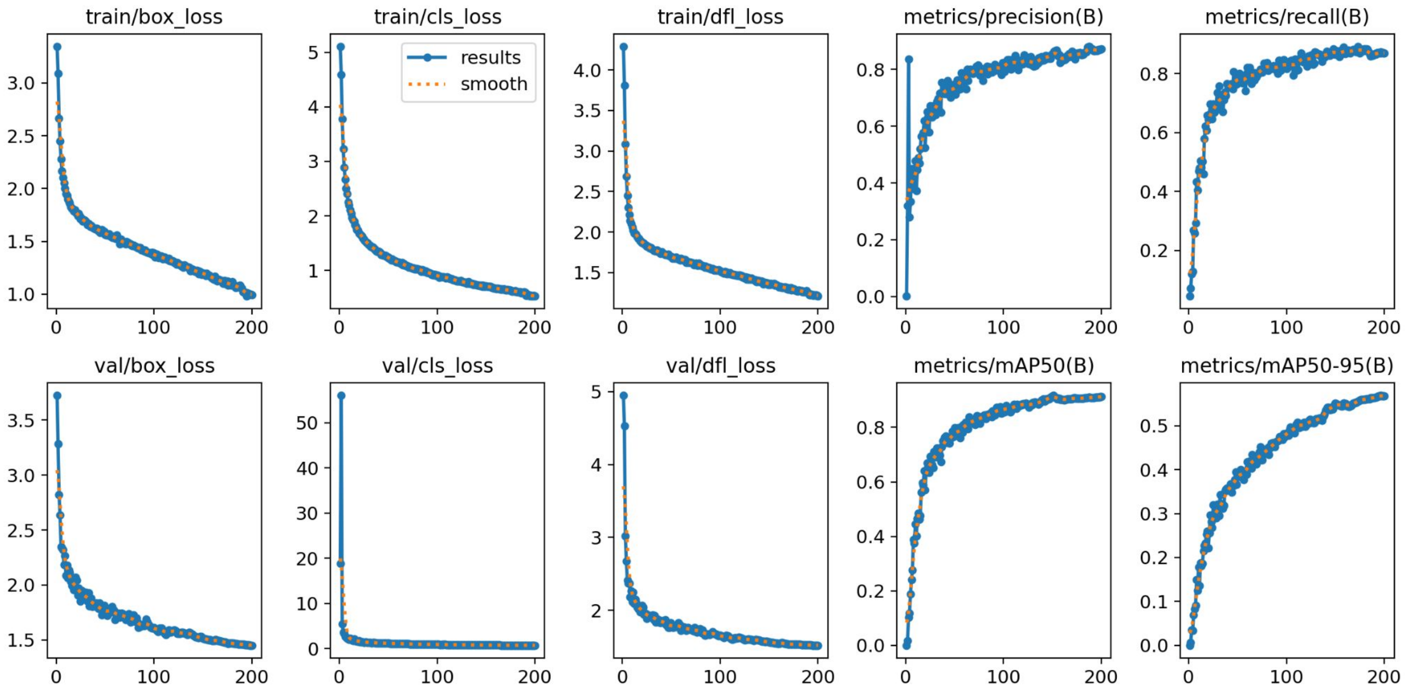
- **YOLO v8.2 -** For object detection.
- **MobileNet-**as shared backbone.

For web development: Streamlit (v1.17)

- Framework for building an interactive web application to deploy models, display results, and capture user feedback.

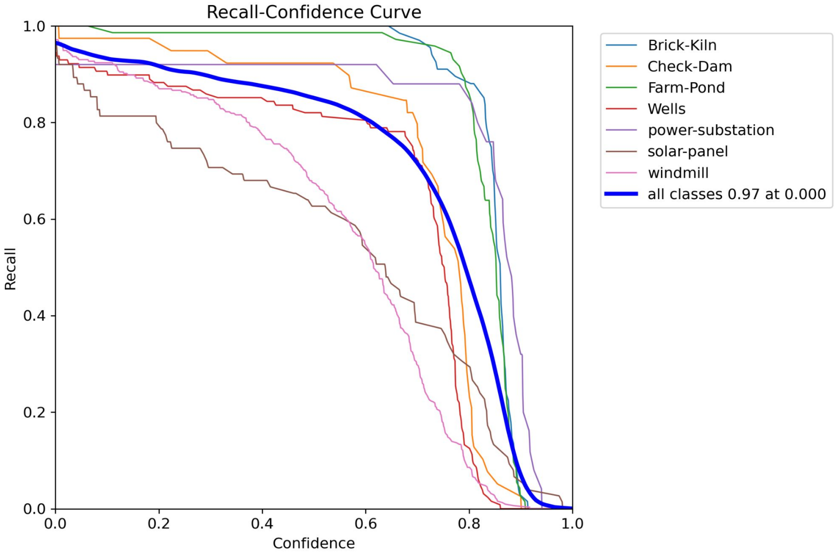
Canny Edge Detection Algorithm : Preprocessing and noise reduction (class specific: farm-bunds)

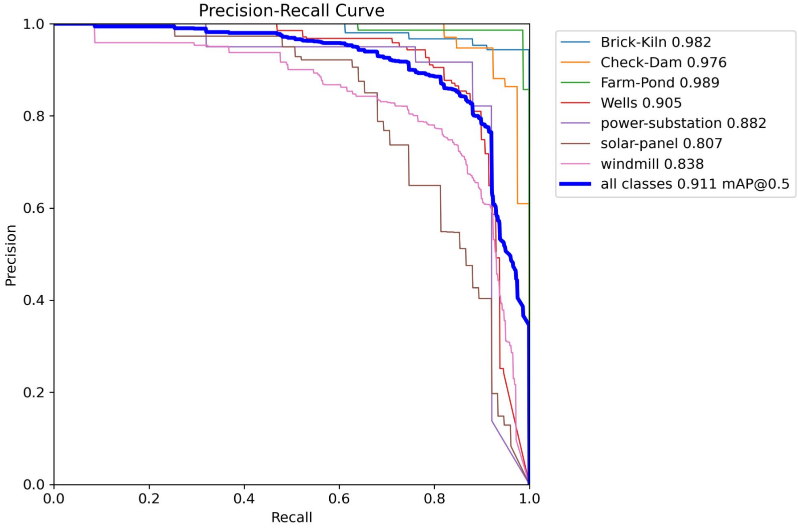




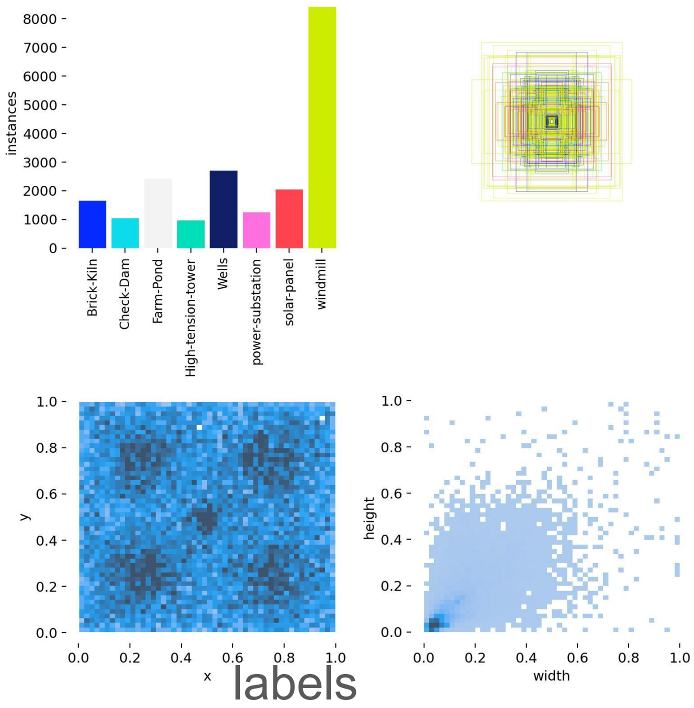
Results from YOLO

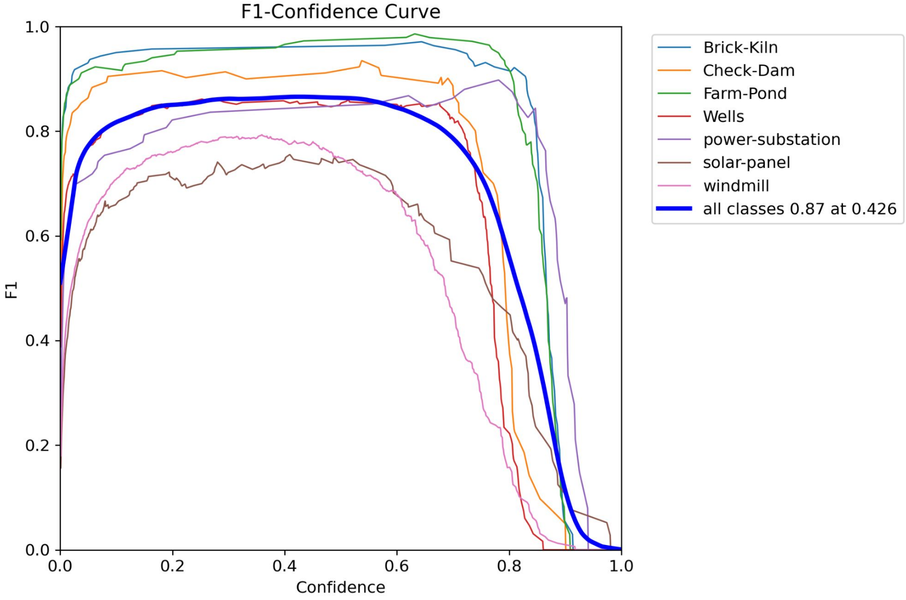
Results from YOLO cont.



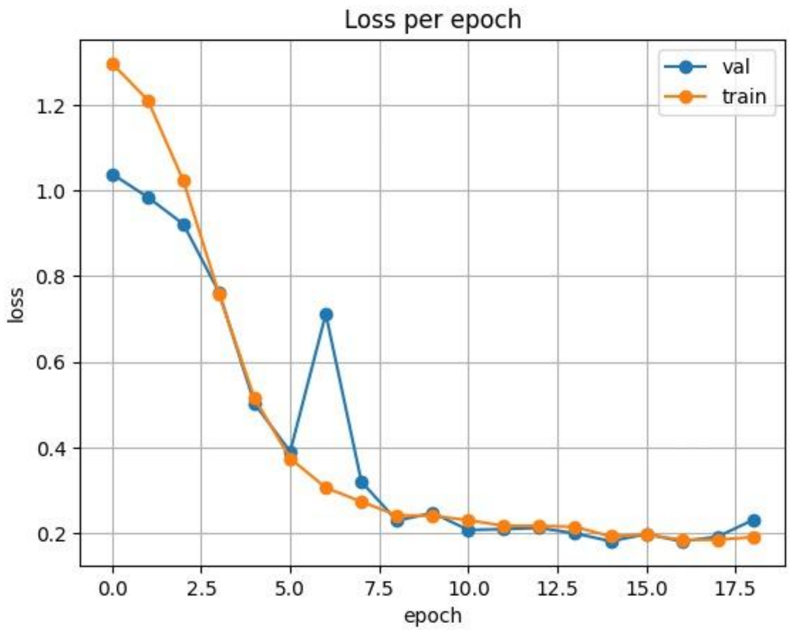


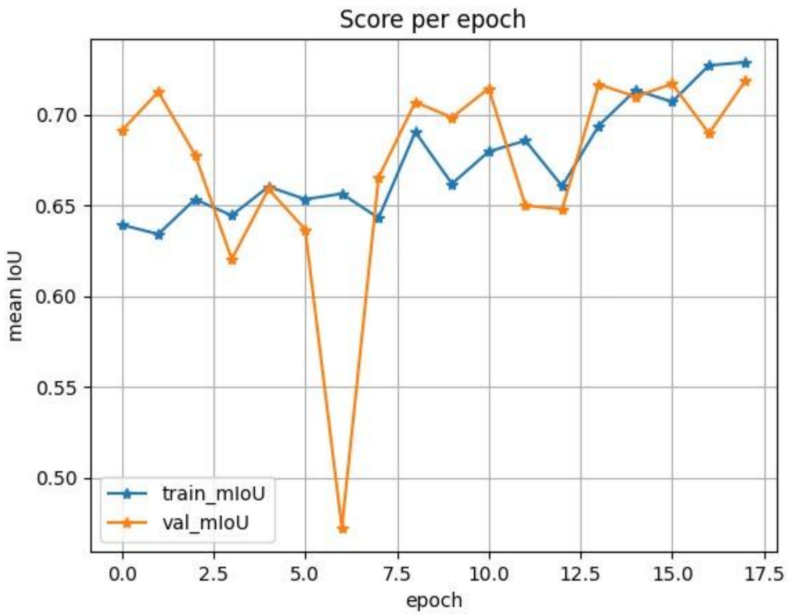
Results from YOLO cont.



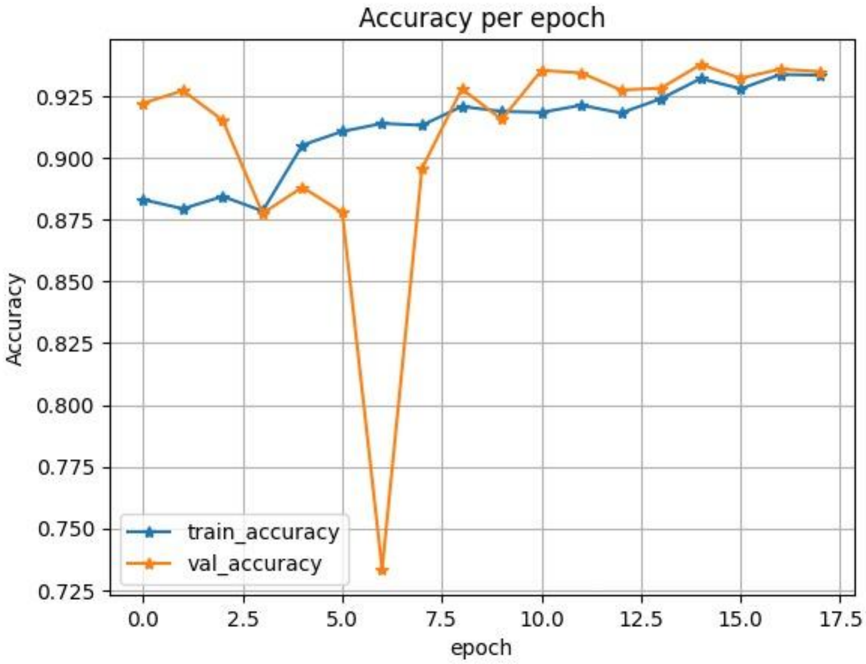


Results from MobileNet

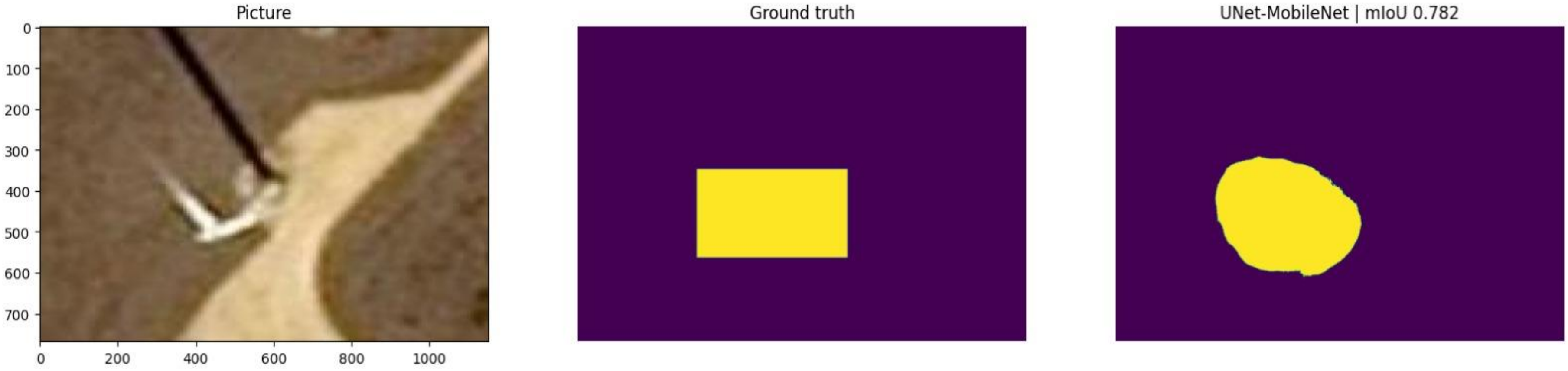




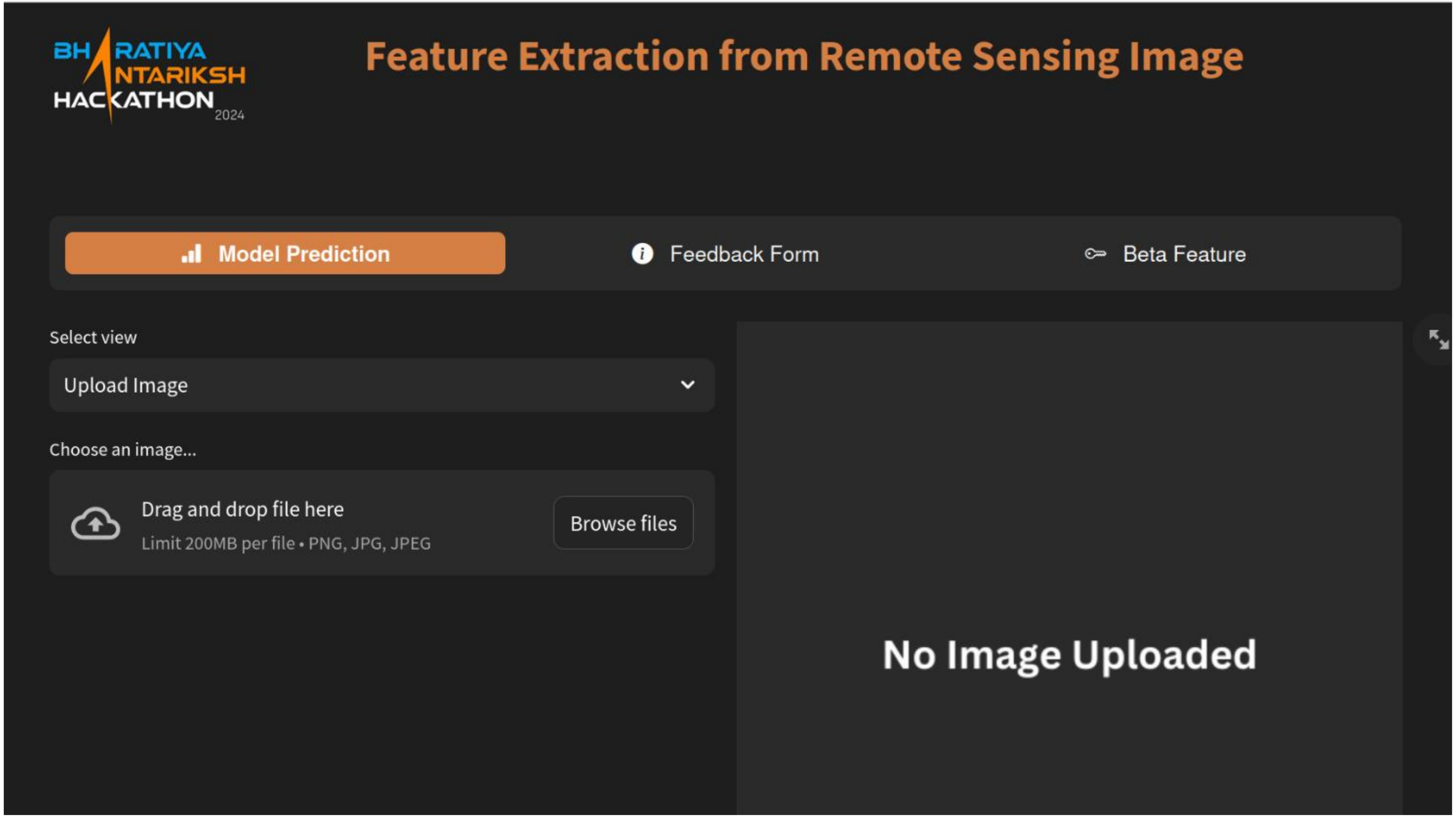
Results from MobileNet cont.



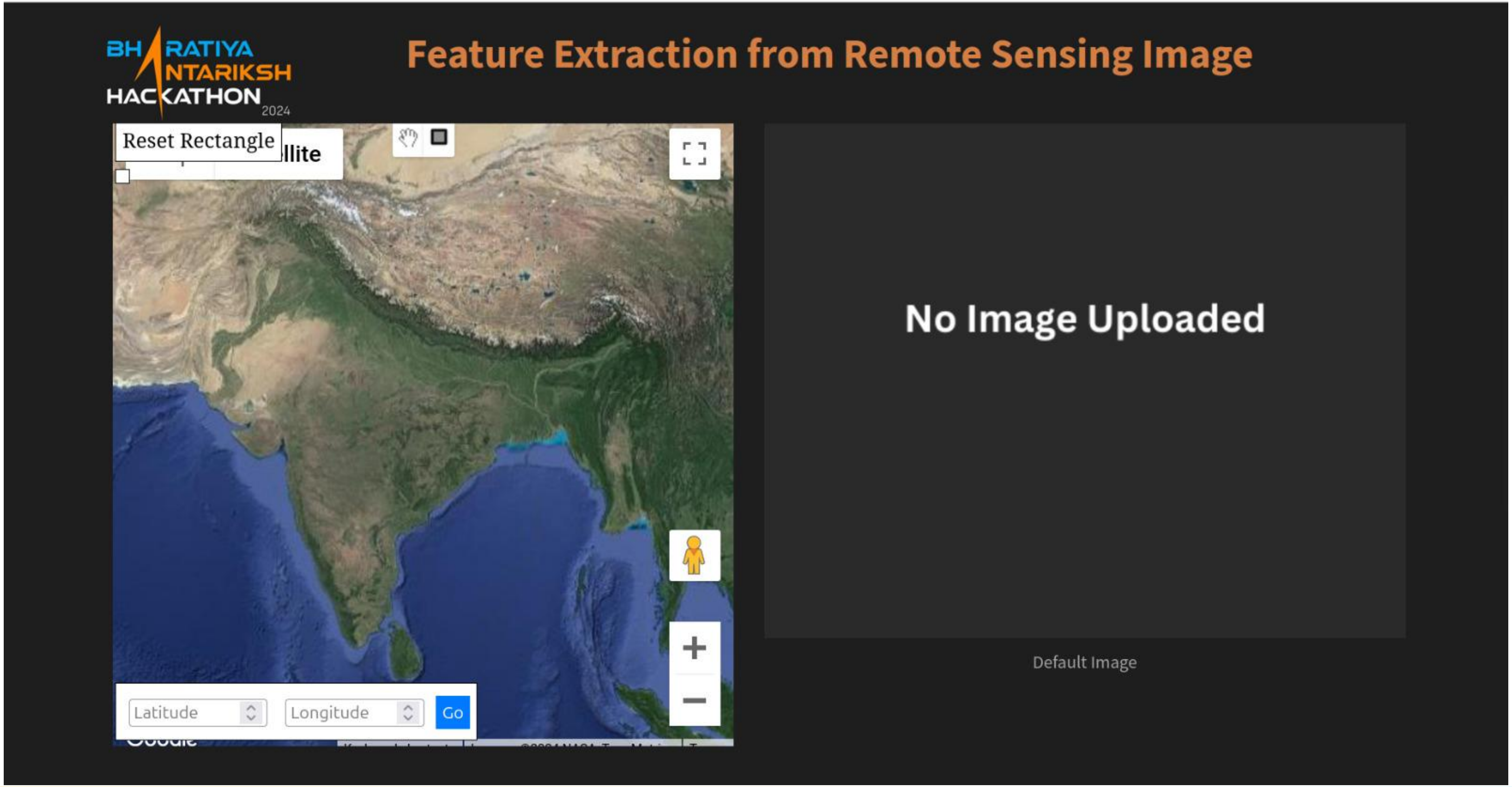
Results from MobileNet cont.



Interactive GUI application Snapshot :Feature extraction using a user input image



On the fly Feature extraction :Using Google API



Feature Extraction from Remote Sensing Image

Drag and drop file here

Limit 200MB per file • PNG, JPG, JPEG

Browse files

Object Detection Dataset 4432.jpg 48.0KB

High-tension-tower

Dataset References

1. Farm-Ponds / Check-Dam / Wells

- **Source**: GramDrishti (IIT Bombay)
- **Image Size**: 640x640 pixels
- **Coverage**: Approximately one hectare of land
- **Regions**: Maharashtra, Madhya Pradesh, Karnataka
- **Source**: Google Static Maps (zoom level 19)
- **References**:
  - https://www.cse.iitb.ac.in/gramdrishti/about/
  - https://arxiv.org/pdf/2111.15613v1 [cs.CV] 30 Nov 2021

2. Transmission Tower

- **Source**: China University of Geosciences (2022)
- **Image Width**: 1000 to 4800 pixels

- **Resolution**: Approximately 0.15 meters
- **Type**: UAV images from top
- **References**:
    - https://github.com/weihancug/10-category-UAV-small-weak-object-detection-dataset-UAVOD10
    - https://datasetninja.com/uav-small-object-detection

3. Farm-Bunds

- **Regions**: Thailand, Indonesia, and India
- **Resolution**: 50 cm/pixel
- **References**:
    - https://www.kaggle.com/datasets/balraj98/deepglobe-land-cover-classification-dataset?select=test
    - https://arxiv.org/pdf/1805.06561
    - https://competitions.codalab.org/competitions/18468#learn_the_details

4. Power-Substation

- **References**:
    - https://www.kaggle.com/datasets/samvram/electrical-substation-detection
    - https://ietcint.com/Ietcint2021/electrical_substation_detection

5. Windmill

- **Source**: Airbus Dataset (SPOT6 and SPOT7 satellites)
- **Resolution**: 1.5 meters per pixel
- **Image Size**: 128 x 128 pixels (≈192 meters on the ground)
- **Reference**: https://www.kaggle.com/datasets/airbusgeo/airbus-wind-turbines-patches

6. Solar-Panels

- **Source**: Deep solar GitHub
- https://github.com/wangzhecheng/DeepSolar

7. Brick-Kiln

- **Source**: Aditi Agarwal repo
- **Coverage**: Indo-Gangetic region (0.4 million square kilometers in India)
- **Identified**: More than 700 new brick kilns
- **References**:
  - https://www.researchgate.net/publication/380263920_Towards_Scalable_Identification_of_Brick_Kilns_from_Satellite_Imagery_with_Active_Learning
  - https://github.com/aditiagarwal-02/brick-kiln

References and Citations:

- https://www.researchgate.net/publication/339786747_Segmentation_of_Satellite_Imagery_using_U-Net_Models_for_Land_Cover_Classification

- https://www.researchgate.net/publication/329739239_Image_segmentation_and_classification_of_large_scale_satellite_imagery_for_land_use_A_review_of_the_state_of_the_arts

- https://www.researchgate.net/publication/379599656_Transfer_Learning_for_Object_Detection_in_Remote_Sensing_Images_with_YOLO

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